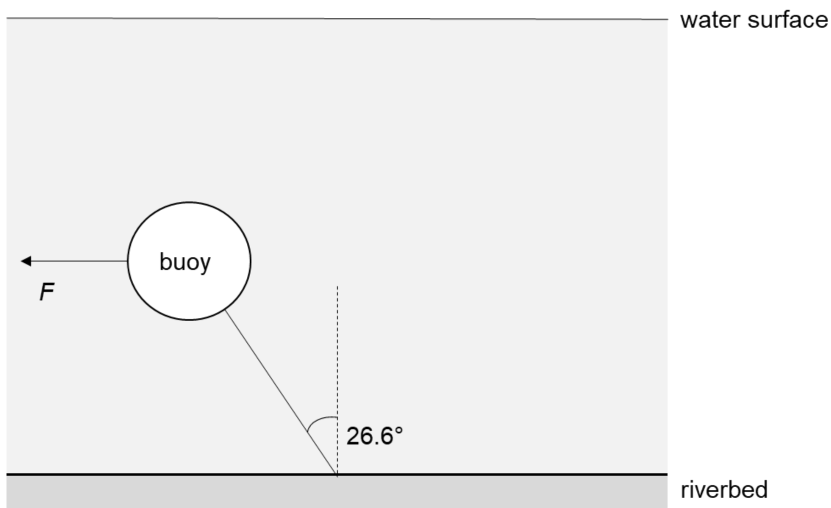
5	A fully submerged buoy is tethered by a rope to a riverbed. The current of the river exerts a constant horizontal force of *F* on the buoy causing the rope to make an angle of 26.6° to the vertical, as shown. The buoy is in equilibrium.



The buoy has a volume of 0.500 m^3 and a density of 390 kg m^{-3}.

The river water has a density of 1000 kg m^{-3}.

What is the force *F* exerted by the river current on the buoy *F*?

A	1500 N	**B**	3830 N	**C**	6000 N	**D**	9810 N

Solution: A

The vertical forces on the buoy are the vertical component of tension, upwards upthrust and the downwards weight.

The net force in the vertical direction is:

$$F_y = U - W$$
$$= V\rho_w g - mg$$
$$= V\rho_w g - V\rho_b g$$
$$= Vg(\rho_w - \rho_b)$$
$$= (0.500)(9.81)(1000 - 390)$$
$$= 3000\text{N}$$

The tangent of the angle that the rope makes with the vertical can be expressed as the ratio of the magnitude of the resultant horizontal force acting on the buoy to the resultant vertical force:

$$\tan 26.6 = \frac{F_x}{F_y}$$

$$0.5 = \frac{F_x}{3000}$$

$$F_x = 1500\text{N}$$

6	An object moves along a frictionless track from A to C, through B, which is the lowest point of